

Symposium

No Central Executive? Decision Formation Through Multi-Area Population Dynamics

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Perceptual decision-making is the process by which sensory evidence is combined with prior knowledge and transformed into possible movement plans according to a rule or policy. Classic studies suggested that perceptual decisions emerge from a feedforward hierarchy of brain areas with distinct functions and fairly homogeneous neural representations. However, more recent findings argue that decisions emerge from distributed, recurrent computations across many brain areas (a “heterarchy”) with complex, heterogeneous representations. How can we make sense of these findings in a way that preserves the computational elegance of the conventional view? In this review, we describe how a new generation of studies is leveraging high-density electrophysiology, incisive task designs, causal manipulations (e.g., optogenetics) and statistical approaches for probing inter-area communication, and theoretical methods that connect population dynamics with representational geometry to build a modern framework for understanding perceptual decisions.

Introduction

The neurobiology of decision-making emerged in the late 20th century as a successful program for establishing quantitative links between neural activity and behavior. The conceptual foundation of this early work was grounded in psychophysics and a “Barlowian” view of neural computation (Barlow, 1972; Parker and Newsome, 1998). The dividend was a satisfying extension of the principles of sensory information processing to a well-defined cognitive function, with single-neuron granularity. However, the price of this elegance was the widespread acceptance of a modular, feedforward perspective that limited investigation to a handful of brain areas and a focus on neurons with simple, interpretable responses. For some, this perspective was consistent with the popular notion of a “central executive,” the hypothetical role of a specific brain region or small set of regions (most notably the prefrontal cortex) that orchestrates the decision process by allocating attention, storing evidence in working memory, applying rules, and serving as input to motor circuitry.

There were early hints casting doubt on this notion. Similar patterns of decision-related activity were observed in multiple premotor and motor structures (Kim and Shadlen, 1999; Romo et al., 2004; Ding and Gold, 2013; Thura and Cisek, 2014; Coallier et al., 2015; Basso and May, 2017), suggesting a parallel, embodied architecture (Shadlen et al., 2008; Cisek and Pastor-

Bernier, 2014) rather than a serial process where a central executive computes the decision and then issues commands to the motor system. But recent work has taken this a step further: as large-scale recording and imaging technologies have become more common, researchers have identified decision-related signals in a surprisingly large fraction of the entire mammalian brain (Musall et al., 2019; Pinto et al., 2019; Steinmetz et al., 2019; Bondy et al., 2024; Khilkevich et al., 2024). Superficially at least, it appears that “everything is everywhere,” the antithesis of a central executive. How do we reconcile the modularity of function from prior studies and anatomy with this heterarchical recurrent system involved in perceiving and deciding? And does the ubiquity of decision signals mean that there is little to be gained by studying individual neurons or brain areas in isolation?

In this review, we highlight how several new interdisciplinary studies—only a small slice of the exciting work being done in this area—are addressing these thorny questions and deepening our understanding of how networks of brain areas implement perceptual decisions. We describe findings from our recent work that uses high-density electrophysiological recordings across multiple brain areas, tasks designed to carefully isolate different cognitive processes, analytical methods that model latent neural dynamics and geometry within and between brain areas, and perturbation techniques that can test how different brain areas influence each other.

Brain-wide recordings suggest decisions emerge from a distributed recurrent system

Technological limitations meant that most early studies of decision-making relied on recordings from a small number of

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brain regions (typically just one), offering an incomplete view of how decision computations unfold across the brain. This limited perspective meant that few experimentalists even considered the possibility that decision computations would be so widely distributed, and what it might mean if they were. The broad adoption of reliable high-density silicon probes (Jun et al., 2017) and cellular Ca^{2+} imaging (Clough and Chen, 2019), has now brought such questions to the fore.

The International Brain Laboratory (IBL) collected brain-wide Neuropixels recordings in mice performing a standardized visual decision-making task (Findling et al., 2025; Laboratory et al., 2025). Across 267 sessions, IBL investigators recorded over 300,000 neurons from 242 brain regions, spanning cortex, midbrain, and hindbrain structures, while mice made left/right choices based on visual stimuli presented with a probabilistic block structure. This massive dataset enabled the IBL to identify the spatial and temporal organization of neural representations of distinct internal and external variables—stimulus, choice, movement, rewards, and prior expectations—on a brain-wide scale.

The data revealed that different task variables were represented with distinct spatial and temporal signatures across the brain (Findling et al., 2025; Laboratory et al., 2025). As expected, visual stimuli evoked rapid, transient responses in the visual cortex and associated thalamic areas. As the trial progressed, these responses evolved into ramp-like activity that extended into the midbrain and hindbrain, revealing a broader pattern of stimulus-driven dynamics. In contrast, choice-related signals emerged more gradually, were sustained over time, and were broadly distributed across many cortical and subcortical regions, including frontal cortex, striatum, and midbrain. Motor-related activity appeared widespread and anticipatory, preceding overt movement by several hundred milliseconds. Prior expectations were manipulated by varying the presentation probability of the two stimulus classes in separate blocks of trials. The prior expectation modulated baseline neural activity and choice representations across many regions, including the motor cortex, frontal cortex, and the basal ganglia. This widespread influence of the prior on neural responses is consistent with a model of Bayesian inference involving loops between areas, as opposed to a model in which the prior is incorporated only in decision-making areas (Findling et al., 2025).

The IBL findings argue against a single centralized executive and instead imply that decision formation emerges through coordinated, dynamic activity across distributed neural populations, broadly consistent with several other rodent studies (Musall et al., 2019; Pinto et al., 2019; Steinmetz et al., 2019; Bondy et al., 2024; Khilkevich et al., 2024). However, despite the widespread nature of these signals, the results show that decision-related information is not represented identically and simultaneously across most of the brain, but rather follows a modular distributed organization: stimulus and choice signals are encoded sparsely in specific regions such as visual and frontal cortex as well as midbrain and hindbrain, while movement and reward signals are observed broadly across nearly all recorded areas.

Ultimately we want to understand not only the “where” and “when” of these categorical representations in the brain, but also how they support specific components of the decision process, a goal that may benefit from more complex tasks and larger-brained species. In the next two sections we summarize work in nonhuman primates aimed at (1) parsing the roles of distinct movement-related neuronal populations in deliberation, urgency, and commitment, and (2) decoupling perceptual judgments from motor planning.

Manipulating the temporal structure of the task reveals how deliberation and commitment unfold in a distributed network

The data from the IBL and other studies provide strong evidence that a remarkable number of brain areas are involved in perceptual decisions. However, in the IBL task, mice viewed a static oriented grating and indicated their response fairly rapidly, with minimal time for deliberation. Expanding the time scale of a decision while encouraging temporal flexibility can provide a deeper understanding of how the brain transitions from a state of deliberation about the evidence for or against a given choice to a state of commitment to a distinct action (Thura and Cisek, 2014; Kira et al., 2015; Stine et al., 2023).

Cisek and colleagues examined how this transition from deliberation to commitment unfolds in the cerebral cortex and basal ganglia of monkeys deciding between two reaching actions on the basis of sensory information that is continuously changing over time (Thura and Cisek, 2014, 2016, 2017; Thura et al., 2022). During the “tokens task” (Fig. 1A), a set of 15 tokens jump, one-by-one every 200 ms, from a central target to one of two potential reach targets; the task is to guess which target will receive the majority of tokens at the end of the trial. Importantly, the monkeys do not have to wait for all of the tokens to jump but can take an early guess, at which point the remaining tokens speed up (to jumping every 150 ms in a “slow” block or every 50 ms in a “fast” block). Consequently, the monkeys are faced with a trade-off between taking quick-and-risky guesses versus making longer-and-safer choices, but the moment of commitment is entirely up to them.

While the monkeys were deliberating about their choice, neural activity in dorsolateral prefrontal (dlPFC), dorsal premotor (PMd), and primary motor cortex (M1) continuously reflected the sensory evidence provided by the token jumps (Fig. 1B). In contrast, neural activity in the globus pallidus (GP) was completely insensitive to the evidence and instead showed an evidence-independent ramping signal that shifted to reflect the change in policy from conservative decisions in the slow block to hasty decisions in the fast block. Around the moment of commitment, these neural activity patterns changed dramatically (Fig. 1C): 280 ms before movement initiation, a prominent peak was observed among PMd cells tuned to the selected target, and afterwards PMd activity became insensitive to the remaining token jumps. Around the same time, activity in GP became selective to the chosen target, and reached a peak of activity about 140 ms later, about the same time as a peak was observed in M1. Meanwhile, dlPFC continued to reflect remaining token jumps.

These results suggest that, at least in the case of decisions between concrete action choices, the decision unfolds as a biased competition between groups of action-selective cells in PMd and M1 (as well as other regions that were not recorded, including medial intraparietal cortex; De Lafuente et al., 2015), which compete against each other until one group wins and suppresses the other (Fig. 1D). This competition is biased by a signal related to evidence that continuously arrives from dlPFC, as well as a choice-independent “urgency” signal from the basal ganglia that pushes the system toward commitment. As the PMd/M1 system reaches a critical contrast between “winning” and “losing” cells, selectivity emerges in the corticostriatal loop, creating positive feedback that constitutes the volitional commitment to a choice (Thura et al., 2022). After that moment, PMd/M1 contribute to the execution of the chosen action while dlPFC continues to monitor task outcomes, potentially shifting baseline activity after errors to adjust the animal’s speed-accuracy trade-off from trial to trial (Thura et al., 2017). In short, decisions about actions unfold

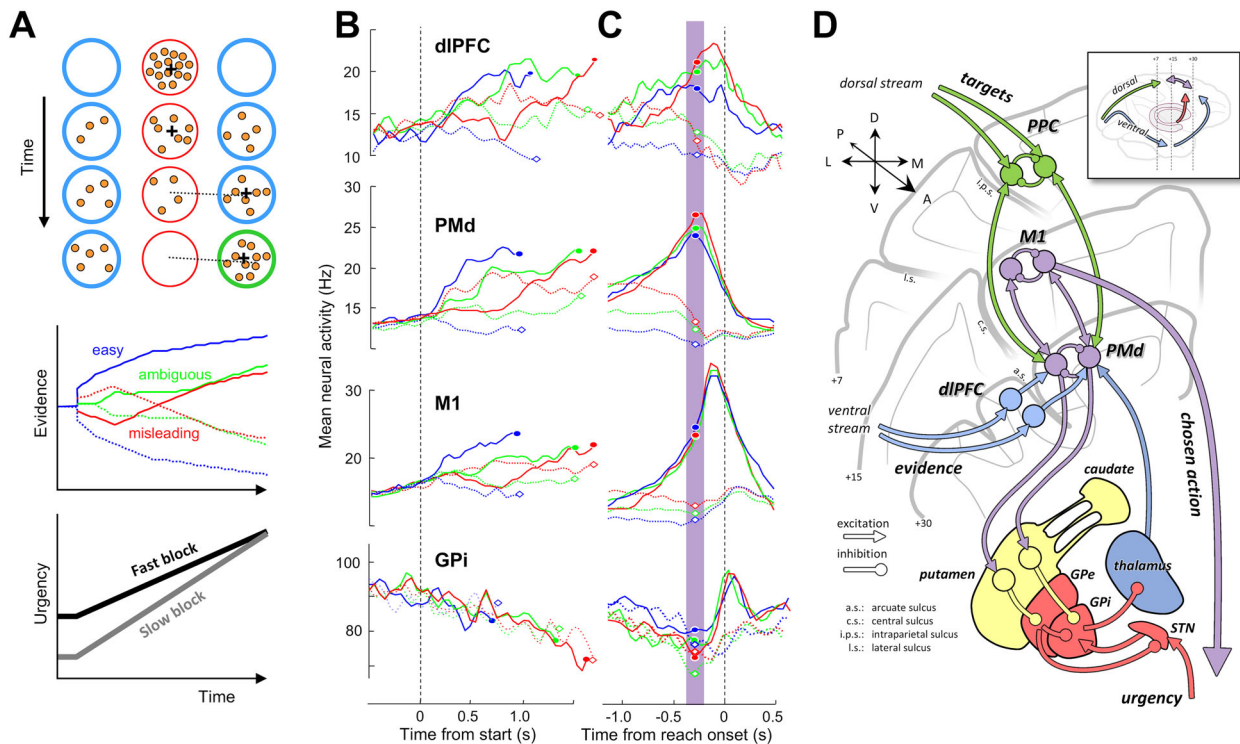


Figure 1. Neural activity during dynamic decision-making. **A**, The “tokens task.” Top: Four frames showing the stimulus display. Black cross indicates the cursor, which the monkey moves into the right target in the third frame, receiving feedback in the fourth after the remaining tokens have jumped. Middle: Temporal profiles of the evidence provided by token jumps in “easy” (blue), “ambiguous” (green), and “misleading” trials (red) for one target (solid) or the other (dotted). Bottom: Temporal profiles of the urgency signal, estimated from behavior, in the “slow block” (gray) and “fast block” (black). **B**, Neural activity in dIPFC, PMd, M1, and GPi, during easy, ambiguous, and misleading trials averaged across cells selective for the chosen (solid) or unchosen target (dotted), aligned on the start of the trial (dashed vertical line). Here, the GPi population only includes “decreasing” cells. **C**, Same activity aligned to movement onset (dashed vertical line). The purple bar estimates the moment of commitment. **D**, A schematic model suggesting how spatial information from the dorsal stream produces competing activity in frontoparietal cortex (PPC, PMd, and M1), which is biased by evidence information from the ventral stream and prefrontal regions, and gradually disinhibited by an urgency signal through the basal ganglia. Thura and Cisek (2017) suggested that once the competition in PMd is sufficiently strongly biased in favor of one action over the other, selective activity emerges in the GPi (see panel **C**, bottom) and completes a positive feedback loop that implements volitional commitment to the choice. Abbreviations: A, anterior; D, dorsal; dIPFC, dorsolateral prefrontal cortex; GPe, globus pallidus external segment; GPi, globus pallidus internal segment; L, lateral; M, medial; M1, primary motor cortex; P, posterior; PMd, dorsal premotor cortex; PPC, posterior parietal cortex; STN, subthalamic nucleus; V, ventral. Panels **B** and **C** are based on data from Thura and Cisek (2017) and Thura et al. (2022).

through a distributed cortico-subcortical circuit that is involved in the control of those same actions, while information about values, risks, effort, and other decision-relevant factors biases the competition until a winner emerges—just as predicted by “recurrent dynamical attractor” models (Thura et al., 2022).

Decoupling perceptual judgments from motor planning can provide insight into the role of different brain areas

In both tasks described above, there is a direct, stable mapping between the sensory evidence and associated actions. In the IBL task, stimuli on the left side of the screen necessitate a turn of the wheel to the right to bring the stimulus to the center, and vice versa. In the tokens task, tokens jump to the left or the right and the animals report their decision by reaching to the corresponding direction. Thus, in these and many other tasks, including the auditory clicks task (Brunton et al., 2013) and most versions of the random-dot motion task (Shadlen and Newsome, 2001), a spatial motor plan can (and does) evolve in tandem with the formation of the decision. Of course, humans and animals regularly make perceptual judgments in the absence of direct motor contingency (Okazawa and Kiani, 2023), but how and where this occurs in the brain is difficult to decipher using these standard task designs.

Wang et al. (2022) were inspired by previous studies (Horwitz et al., 2004; Bennur and Gold, 2011; Freedman and Assad, 2011;

Coallier et al., 2015; Shushruth et al., 2022; Charlton and Goris, 2024) that varied and/or delayed knowledge of the mapping between evidence and actions. The goal was to separate perceptual processing from motor planning, and thereby yield new insight into the dynamics in different brain areas during perceptual decision-making. Wang et al. (2022) trained monkeys to discriminate the dominant color of a central checkerboard and report their decisions with an arm movement. By randomizing target configurations on a trial-by-trial basis, the authors were able to unlink the perceptual judgment (color of the checkerboard) from the action choice (left vs right). They used Neuropixels and laminar multi-contact probes to record from populations of neurons in PMd and dIPFC while animals performed the task.

Based on previous demonstrations of decision-related dynamics in PMd (Thura and Cisek, 2014; Coallier et al., 2015; Chandrasekaran et al., 2017; Boucher et al., 2023) and reports of decision signals in dLPFC (Kim and Shadlen, 1999), Wang and colleagues expected to find target configuration, color choice, and action choice signals in both PMd and dIPFC. Instead, neural dynamics in dIPFC strongly covaried with target configuration, color choice, and action choice (Wang et al., 2022), whereas activity in PMd only covaried with the action choice (Wang et al., 2019). These results were consistent with multi-area recurrent neural network (RNNs) models of decision-making that

predicted stronger choice signals in areas closer to the output, whereas contextual signals are stronger in earlier areas (Kleinman et al., 2025). These results suggest that decoupling perception from actions can help isolate the neural dynamics underlying distinct components of a perceptual decision. A similar conclusion was reached by Charlton and Goris (2024), who observed an initial phase of abstract deliberation in PFC followed by emergence of the concrete motor plan.

Importantly, these results should not be interpreted as evidence that PMd is simply a readout module in a feedforward network. In another study, Boucher and colleagues (Boucher et al., 2023) noticed that reaction times (RTs) in the checkerboard decision-making task are quite variable even for easy discriminations (e.g., a near fully red checkerboard). Consistent with this behavioral variability, the pre-stimulus state in PMd covaried with RT and was altered by the outcome of the previous trial. These findings are in good agreement with observations that speed-accuracy trade-offs alter the baseline firing rates of PMd (Thura and Cisek, 2016) and frontal eye field (FEF) neurons (Heitz and Schall, 2012).

Projection-specific optogenetics combined with computational modeling can reveal how brain areas coordinate their activity during decisions

Population recordings from multiple areas can tell us that decision computations are widely distributed. However, recordings alone cannot directly probe the circuit mechanisms of inter-area coordination, or the contributions of distinct cell types. Importantly, if the goal is to reveal how one area's activity drives another, neural activity measurements are generally insufficient, despite the long history of inferring causal interactions from time series data in cognitive neuroscience (discussed extensively in Mehler and Kording, 2018; Fakhar et al., 2023; Lepperød et al., 2023). This is because brain areas are heavily interconnected through recurrent loops, with any given area simultaneously acting as both the cause and effect of activity in connected areas. Furthermore, recorded areas may receive common input from unobserved areas, giving rise to correlated activity even in the absence of direct causal interactions. Therefore, although recent efforts have introduced a variety of statistical tools to infer causal structure among simultaneously recorded brain areas (e.g., "communication subspace" and other multivariate regression and dimensionality reduction methods; Kohn et al., 2020; Semedo et al., 2020; Gokcen et al., 2022), correlative measures of neural activity remain inherently ambiguous proxies for causal inter-area interactions (Das and Fiete, 2020). Perturbing distinct circuit components, i.e., with opto- or chemogenetics, can help resolve these ambiguities, as exemplified by recent studies (Gupta et al., 2024; Javadzadeh et al., 2024) that combined multi-area recordings, projection-specific perturbations, and modeling (Fig. 2).

Previous work applying optogenetics in rats during an auditory evidence accumulation task suggested a modular, feedforward, and hierarchical architecture for perceptual decisions: evidence accumulation was localized to anterior dorsal striatum (Yartsev et al., 2018), while decision commitment was associated with specific parts of frontal cortex [the frontal orienting field (FOF); Erlich et al., 2015; Hanks et al., 2015]. In contrast, Gupta et al. (2024) found surprisingly strong evidence against this simple modular and feedforward view and instead provide evidence for a heterarchy. Simultaneous recordings revealed that the striatum and FOF have indistinguishable decision-related activity, showing similar magnitudes and timescales of evidence encoding/decoding, and the two regions share this

information with each other in their communication subspace. More counterintuitively, projection-specific inactivations revealed a complicated relationship between perturbations and function. While inactivating the entire FOF had no effect on behavior during evidence accumulation, specifically targeting and inactivating FOF inputs to striatum did have an effect, implicating the frontal cortex in evidence accumulation (schematized in Fig. 2A,B).

To understand these puzzling results, the investigators modeled the interactions between frontal cortex and striatum in two ways, representing different approaches to understanding multi-region circuit function. First, they used a perturbation-based approach, training anatomically constrained multi-region RNNs to reproduce behavioral patterns on both control and optogenetic inactivation trials (Fig. 2A,B). This novel method trained networks to match the functional consequences of circuit manipulations rather than neural activity patterns per se. The results revealed that inactivation of the frontal cortex fails to impair behavior because it is rescued by multiple scales of recurrence in the circuit (self-recurrence, inter-hemispheric recurrence, and striato-cortical recurrence), while frontostriatal projection-specific inactivations are not rescued in the same way. Remarkably, the model reproduced extensive features of decision-related representations in the two areas despite not being trained to do so. It correctly predicted effects of unseen perturbations and generated counterintuitive predictions. For example, it predicted that inactivating cross-hemispheric FOF projections in addition to corticostriatal ones should paradoxically eliminate decision impairments caused by inactivation of the striatum-projecting FOF neurons. This suggests that perturbation-based constraints can capture fundamental circuit-level neural principles. Second, they took a complementary dynamics-based approach (Jha et al., 2024) and fit multi-region dynamical systems models with cell-type specific linear dynamics. Intriguingly, this model—trained purely on neural dynamics without any exposure to perturbation data—successfully reproduced the empirically observed perturbation effects.

The convergence of these two modeling approaches provides powerful evidence for the distributed, recurrent implementation of evidence accumulation. The perturbation-based approach captures how circuits maintain function under disruption and can incorporate extensive anatomical projection patterns while making novel experimental predictions about complex, nonlinear circuit interactions. Dynamics-based models, while technically challenging to imbue with detailed anatomy or nonlinear dynamics, provide more directly interpretable insights into the computational mechanisms underlying neural activity patterns. Critically, both approaches required explicit multi-region architectures that respect known anatomical connectivity patterns.

Javadzadeh et al. (2024) used a similar strategy to understand multi-area sensory processing in the mouse visual cortex, specifically investigating the interactions between primary visual cortex (V1) and the higher-order lateromedial visual area (LM) during visual perception. In earlier work (Javadzadeh and Hofer, 2022), they measured neural activity in V1 and LM and quantified cross-area interactions using simultaneous electrophysiological recordings and transient optogenetic perturbations (Fig. 2C,D). They found that neurons across these areas continually exchange complex patterns of information that dynamically evolve on fast time scales during the course of visual perception.

To better understand these complex inter-area interactions, the new study combined mechanistic modeling of cortical circuits with data-driven inference of neural dynamics to construct

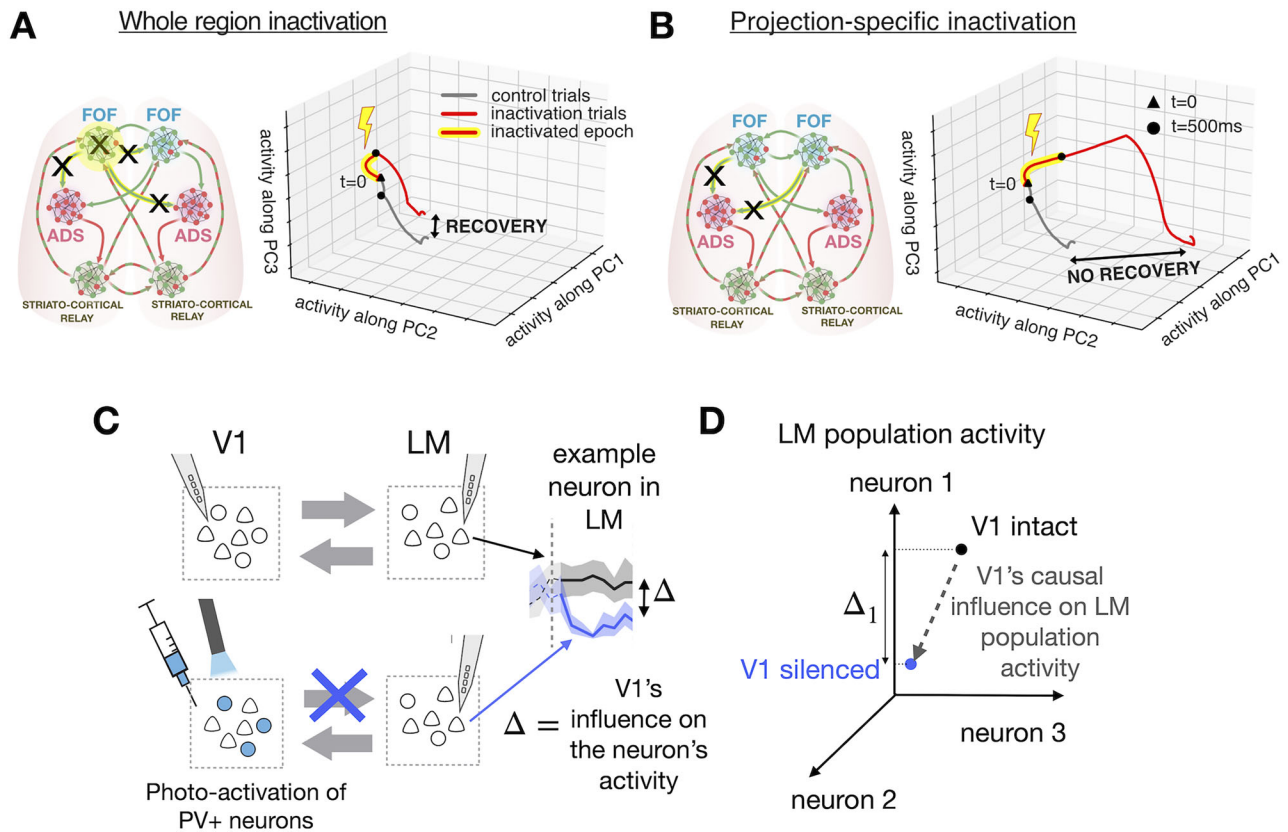


Figure 2. Perturbing neural activity to interrogate multi-area computations **A–B**, Projection-specific optogenetic manipulations reveal distinct compensatory mechanisms in corticostriatal circuits during evidence accumulation, from Gupta et al. (2024). Circuit diagrams show frontal orienting field (FOF), its direct projections to anterior dorsal striatum (ADS), and recurrent connections back via a module representing the multi-synaptic relay between ADS and FOF (i.e., other basal ganglia nuclei, thalamus, superior colliculus, etc.). 3D plots show low-dimensional projections of population activity along the principal components during control (gray) and inactivation (red) trials. **A**, Whole region unilateral inactivation: Complete FOF silencing (X marks) during the early evidence accumulation epoch initially perturbs dynamics but allows recovery through multiple compensatory recurrent loops, preserving decision-making behavior. **B**, Projection-specific inactivation: Selective targeting of FOF-to-ADS projections (X marks on corticostriatal pathway) during the same epoch produces sustained deficits without recovery, revealing the critical role of corticostriatal communication in evidence accumulation. **C–D**, Using optogenetics to map the signal flow between cortical areas, from Javadzadeh and Hofer (2022). **C**, Top, simultaneous electrophysiological recordings in two anatomically connected visual areas in mice, primary visual cortex (V1) and lateromedial higher visual area (LM), using multi-channel silicon probes, yielding multi-area neural recordings at single-neuron resolution. Bottom, V1 is optogenetically silenced in brief time windows using light-mediated activation of ChR2 expressing PV+ interneurons. Right, for any given recorded neuron in LM, the difference between activity levels in control (black) and V1-silencing (blue) trials indicates V1's influence on it. **D**, In the high dimensional population activity space of LM, comparing the activity in control and V1-silencing trials will give a population-level measure of V1's feedforward influence on LM.

biologically constrained latent variable models (Javadzadeh et al., 2024). This approach allowed them to not only explain neural responses quantitatively, but also capture the effects of optogenetic perturbations. Moreover, the models contain biologically interpretable components—including local and long-range connections—whose functional significance could be interrogated. This method revealed that reciprocal excitatory connections between V1 and LM generate joint dynamics across areas that resemble an approximate line attractor. This attractor selectively sustains activity patterns that encode the presence of the stimulus consistently across the two areas. Further theoretical analyses suggested that such attractor-like dynamics reflect a broader principle of multi-area networks: long-range reciprocal projections between cortical areas actively reshape neural dynamics, specifically sustaining neural activity patterns that encode stimulus features consistently across areas, while accelerating the decay of conflicting ones. This dynamical regime promotes the alignment of activity patterns across areas and leads to the dynamic emergence of a distributed “consensus” across areas. This framework also predicted the timescales of specific cross-area activity patterns, which were subsequently validated in the data. These results point toward a general mechanism by which distributed

processing across cortical areas can be shaped by reciprocal loops between them (Javadzadeh et al., 2024).

Heterogeneous neural populations may conceal common dynamical motifs for decision formation

We have summarized multiple studies that combined optogenetics and statistical tools to identify how neural activity in one area covaries with activity in another area. These approaches will be crucial for dissecting functional interactions between distinct neural populations. Additional insight into the nature of the computations performed by these interconnected populations will also benefit from new analytical approaches that provide a deeper understanding of the dynamics within a given brain area (Kim et al., 2021; Pals et al., 2024; Genkin et al., 2025).

Genkin et al. (2025) used such an approach to better understand how populations of neurons in PMd support perceptual decisions. Conventional theories assume that neurons encode decision variables linearly or monotonically in their firing rates (Schall, 2019). However, this hypothesis seems inconsistent with heterogeneous responses of single neurons in decision-related brain areas, which often show complex temporal response profiles (Mante et al., 2013; Meister et al., 2013; Park et al., 2014;

Chandrasekaran et al., 2017). How can this temporal heterogeneity be reconciled with the population-level encoding of low-dimensional decision variables (DVs)? Consider an analogy to sensory systems. Sensory variables are represented in the coordinated activity of neural populations, where individual neurons' tuning curves collectively define how the population represents information (Kriegeskorte and Wei, 2021). Genkin et al. (2025) tested if the same coding principle could explain decision-related neural responses (Fig. 3). They developed an approach to simultaneously infer population dynamics $[x(t)]$ encoding a latent variable and tuning functions of single neurons $[f(x)]$ to this latent variable. Applied to spiking activity from PMd neurons during the aforementioned checkerboard task, the model revealed that populations of neurons encoded the same dynamic DV predicting choices, and heterogeneous firing rates resulted from the diverse tuning of single neurons to this DV.

One benefit of separating internal dynamical variables associated with decisions from how a given neuron is tuned to these dynamical variables (as opposed to external variables, Christensen et al., 2022) is that it may help resolve apparent conflicts in the empirical literature and build better models of decision-making. For example, many studies of macaque lateral intraparietal area (LIP; Churchland et al., 2011; Kira et al., 2015; Steinemann et al., 2024) emphasized dynamics in which a drift-diffusion-like process is embedded in persistent activity ("ramping"), whereas other studies (especially in rodents) described sequences of activity propagating across neurons (Harvey et al., 2012). Still other studies, including in LIP (Meister et al., 2013; Zoltowski et al., 2019), FEF (Ding and Gold, 2012), and PMd (Chandrasekaran et al., 2017) highlighted

the heterogeneity that encompasses both extremes (see also Bukwich et al., 2024). It is intriguing to speculate that some of the discrepancies between studies may be reconciled by positing diverse neural tuning to a common set of relatively simple dynamical variables. Further development of this approach could entail estimating dynamical variables in multiple dimensions while assuming nonlinear tuning to these variables. The hope is that by gaining better traction on within-region dynamics, the field will be better equipped to understand multi-region computations at a level beyond the mere existence and basic properties of a communication subspace.

DV representations serve other purposes than the decision itself

Thus far we have described approaches to help tame the complexity of multi-region high-density recording and dispel the illusion of "everything is everywhere." While the death of modularity may have been exaggerated, it is worth noting that some true redundancy may exist to support distinct computations occurring in parallel, including capacities that standard decision tasks do not directly test. Consider that a neural representation of a DV is required not only for computing the current choice but for informing the next one in a sequence (Van den Berg et al., 2016; Zylberberg, 2021), or for building internal models to improve future decisions. Indeed, a hallmark of model-based learning (Doll et al., 2012; Lee et al., 2012) is the ability to associate internal states, not just overt actions, with outcomes.

One such internal state is the metacognitive sense of confidence: a prediction of accuracy based on a nonlinear readout or other operation applied to the DV (Kiani et al., 2014; Shekhar and Rahnev, 2021; Boundy-Singer et al., 2023). Confidence is critical for interpreting feedback under changing conditions (Purcell and Kiani, 2016; Sarafyazd and Jazayeri, 2019; Lak et al., 2020) and for sequencing decisions in the absence of feedback (Van den Berg et al., 2016; Zylberberg, 2021), but how this occurs is constrained by where and when confidence is computed in the brain.

Vivar-Lazo and Fetsch (2025) trained monkeys to report their choice and confidence simultaneously with a single eye movement ("peri-decision wagering") in a RT motion discrimination task (Fig. 4A). They found that neural dynamics in area LIP during decision formation evolve concurrently in two dimensions, one for resolving the choice and the other the wager (Fig. 4B). This parallel architecture for choice and confidence, consistent with recent work in humans (Balsdon et al., 2021; Xue et al., 2023; Dou et al., 2024), implies the availability of metacognitive signals in real time for other computations, such as deciding whether to seek further information, revising an ongoing motor plan, or making online adjustments to the accumulation process (Balsdon and Philiastides, 2024). In each case the brain might distribute seemingly redundant decision signals to different sub-networks (Fig. 4C), yet their purpose would remain obscure in the absence of a behavioral requirement to utilize them. This may also explain the lack of Katz et al. (2016), or compensation for Jeurissen et al. (2022), behavioral effects in some reversible inactivation studies.

The key point is that high-order cognitive functions, such as confidence, take lower-order representations (e.g., a DV) as input, just as the primary decision requires operations (pooling, integration, etc.) applied to the sensory data. Because of its utility for guiding flexible behavior in various contexts, we should expect "global broadcast" of confidence and its constituents throughout prefrontal and parietal association cortices

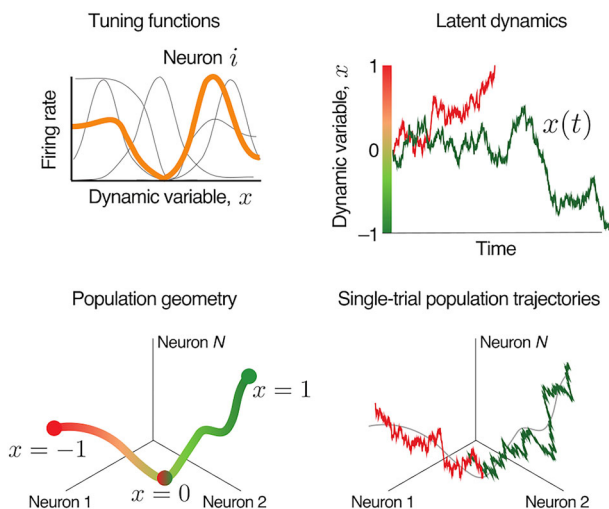


Figure 3. Elucidating decision computations in heterogeneous neural populations: Genkin et al. (2025) propose that neurons are tuned to internally generated dynamic variables that implement cognitive computations (top left). These dynamic variables evolve with distinct time courses on single trials (top right). Tuning functions describe how firing rates depend on the value of the dynamic variable, independent of time in the trial or task condition (top left). These tuning functions uniquely specify the representational geometry of the dynamic variable in the population state space, precisely as they do for sensory stimuli (Kriegeskorte and Wei, 2021). For example, the value of the decision variable at any given time is encoded by the position of population activity along a one-dimensional manifold in the population state space (lower left). The population trajectories traverse this manifold as they evolve toward one or another choice on each trial. The shape of these population trajectories is uniquely specified by tuning functions of single neurons to the decision variable (top left), with the same dual relationship between neural tuning and population geometry as for sensory stimuli. The dynamics of the decision variable describe how population activity progresses along this manifold on single trials (lower right).

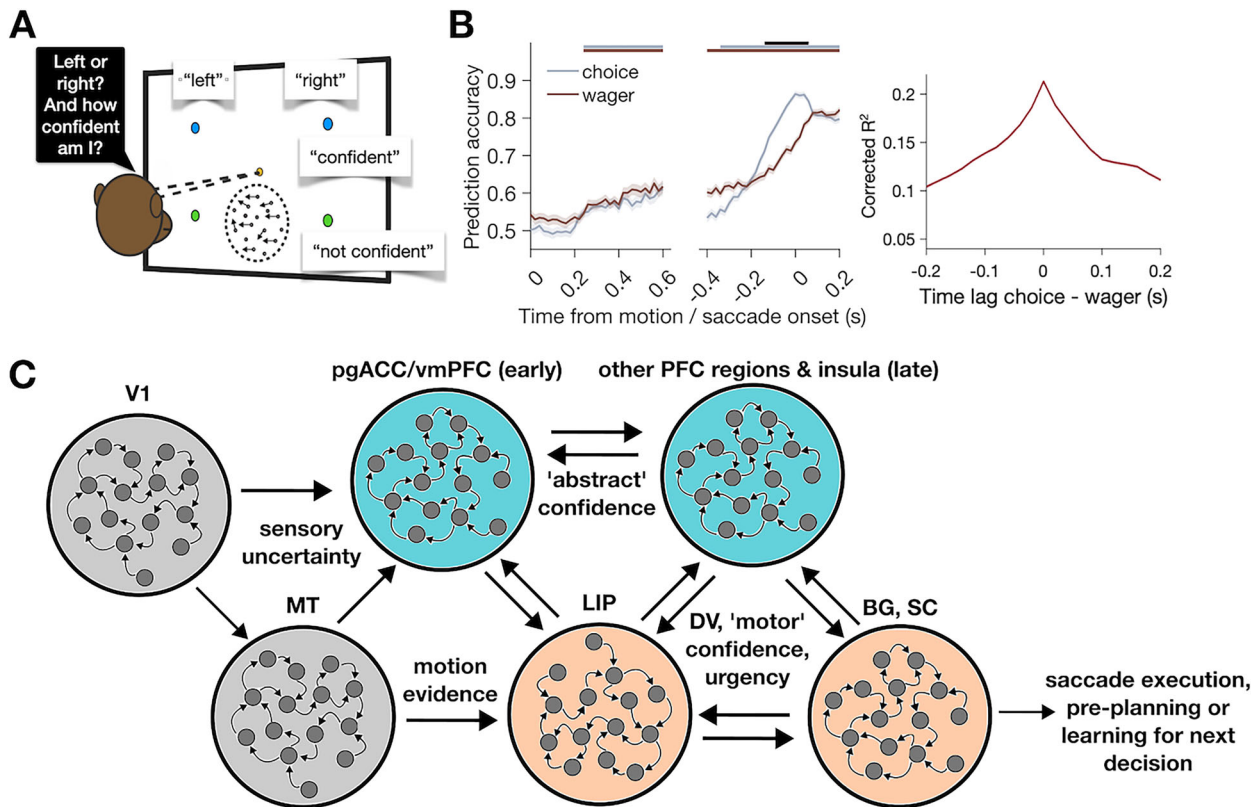


Figure 4. Dynamics of decision confidence and hypothesized brain network for visual metacognition: **A**, Vivar-Lazo and Fetsch (2025) probed the temporal evolution of choice and confidence formation by analyzing saccadic decision signals in posterior parietal area LIP while monkeys performed a “peri-decision wagering” task. The monkey viewed a random-dot stimulus and whenever ready made a saccade to one of four targets, simultaneously indicating a binary direction judgment and a binary wager on the outcome of that decision. **B**, Population decoding in LIP revealed concurrent updating of decision and confidence signals: decoding accuracy for the direction choice and the wager ramped up with a similar time course (*left*), and within-trial fluctuations in the strength of choice and wager encoding covaried with essentially zero time lag (*right*, corrected R^2 from a linear model relating single-trial choice and wager decoder outputs as a function of lag between them; see Vivar-Lazo and Fetsch (2025) for details). **C**, Many brain areas have been implicated in the construction of confidence during and after a decision. There is no consensus “wiring diagram,” but human and monkey studies suggest that parietal association cortex (e.g., macaque LIP) represents a decision variable (DV) from which confidence may be computed “online” in a motor-centric or intentional reference frame. In parallel, a higher-order network [PFC (including FEF), orbitofrontal cortex (OFC, not shown), ACC, insula] queries the DV and other sources to construct a more abstract or domain-general confidence signal, which then updates ongoing activity in sensorimotor, premotor, and basal ganglia circuits to guide action selection, motor vigor, and learning. Abbreviations: V1, primary visual cortex; MT, middle temporal area; pgACC/vmPFC, perigenual anterior cingulate cortex & ventromedial prefrontal cortex (Fleming, 2024); LIP, lateral intraparietal area; BG and SC, basal ganglia and superior colliculus.

(Fleming, 2024) and even subcortically (Komura et al., 2013; Hebart et al., 2016). An important challenge for future work is to relate the dynamics of specific internal states to the flow of information through subsets of this extensive network (Fig. 4C), perhaps aided by targeted perturbation and analytical strategies like those highlighted in this review.

Conclusion and Future Directions

Collectively, these studies have provided four key takeaways. First, it is inescapable that computations for perceptual decisions are distributed and thus the next generation of frameworks should embrace this fact. However, this does not mean that everything is everywhere all at once (Wang et al., 2023). There is instead a systematic, interpretable flow of information and distinct neural dynamics within different nodes of these distributed circuits. The implication is that unstructured RNNs (“bag-of-neurons”) and other computational models that assume complete redundancy are unlikely to be successful in capturing biological circuit mechanisms, and may only provide limited insight into human-like intelligence.

Second, perturbations and projection-targeting approaches that focus on the edges in the network can complement studies

of the nodes, identify directions of information flow, and how dynamics in one area influence the dynamics in another. Making perturbation approaches more accessible across species (including nonhuman primates), and integrating them with correlational techniques and modeling, will be increasingly critical for achieving a mechanistic understanding of this distributed network.

Third, we need thoughtful task designs and new decision-making paradigms that balance complexity with interpretability. An unfortunate consequence of distributed processing is that it makes experiments to assess and infer causation much harder to conduct and interpret (Wolff and Ölveczky, 2018), especially if the task designs are simple, because many different pathways and brain areas could be involved in simple decisions. Thus, modern task designs that produce interpretable behavioral variability in more naturalistic contexts will become even more important.

Fourth and finally, new methods that can better elucidate the link between neural population dynamics and the complex response profiles of individual neurons will be critical for a more refined understanding of modularity. Overall, the payoff of this effort will be to help fill the ‘mesoscale gap’ in neuroscience—our understanding of interacting neural populations lags behind the biology of microcircuits and the cognitive neuroscience of whole-brain networks—with the potential to drive qualitative

improvements both in artificial systems and treatments for disorders that affect cognition.

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